



MATHEMATICS

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write on the lined paper in the booklet provided.
- **Each question is to commence on a new page.**
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- **This paper must NOT be removed from the examination room by the candidate.**

- ☐ Mr Ireland
- ☐ Mr Lam
- ☐ Mr Lowe
- ☐ Mr Fletcher
- ☐ Mr Trenwith/Mr Taylor

STUDENT NUMBER:

[illegible]

Question 1 (12 Marks)

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Marks

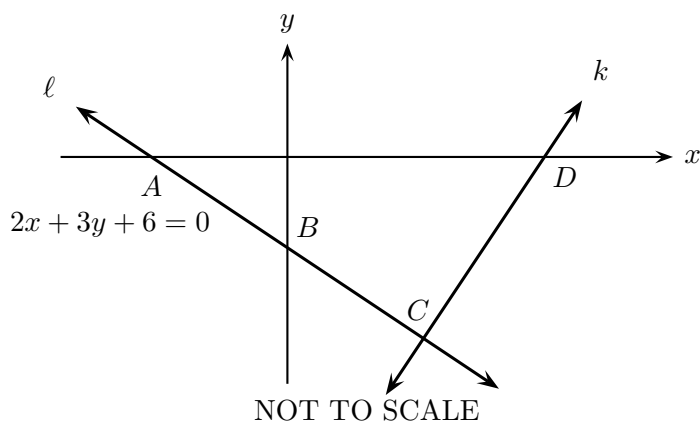
- (a) Evaluate $\left(\frac{1}{e^{2.5}} - 1\right)^2$ correct to 3 significant figures. **2**
- (b) Express $\frac{\sqrt{2}}{1 + \sqrt{5}}$ with a rational denominator. **2**
- (c) Differentiate $y = (4x + 1)^3$ with respect to x . **2**
- (d) Factorise $x^4y - xy^4$ fully. **2**
- (e) Solve the following for x :
- i. $2^{2x-3} = 32$. **2**
- ii. $x^2 - x = 2$. **2**

Question 2 (12 Marks)

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Marks

- (a) The line ℓ has the equation $2x + 3y + 6 = 0$. It cuts the x axis at A and the y axis at B and it intersects the line k at C . Line k is perpendicular to ℓ and cuts the x axis at D .



Copy or trace the diagram on to your paper.

- i. Find the coordinates of A . **1**
 - ii. Find the coordinates of B . **1**
 - iii. If B is the midpoint of AC prove that the coordinates of C are $(3, -4)$. **2**
 - iv. Show that the equation of k is given by $3x - 2y - 17 = 0$. **2**
 - v. Write the 3 inequalities required to define the interior region of $\triangle ACD$. **3**
- (b) Find the equation of the tangent to the curve $y = x^2 \ln x$ at the point P where $x = e$. **3**

Question 3 (12 Marks)

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Marks

- (a) Consider the parabola $(x - 2)^2 = 8(y + 1)$.
- i. Write down the focal length. **1**
 - ii. Write down the coordinates of the focus. **1**
 - iii. Find the equation of the directrix. **1**
- (b) Differentiate with respect to x :
- i. $2x^3 - x^{-1}$. **2**
 - ii. $\frac{\sin x}{e^{2x}}$. **2**
- (c) Evaluate $\int_1^e \left(x^2 + \frac{2}{x}\right) dx$. **2**
- (d) Find an approximation for $\int_1^3 g(x) dx$ by using Simpson's Rule with the following function values in the table below, correct to 2 decimal places. **3**

x	1	1.5	2	2.5	3
$g(x)$	12	8	0	3	5

Question 4 (12 Marks)

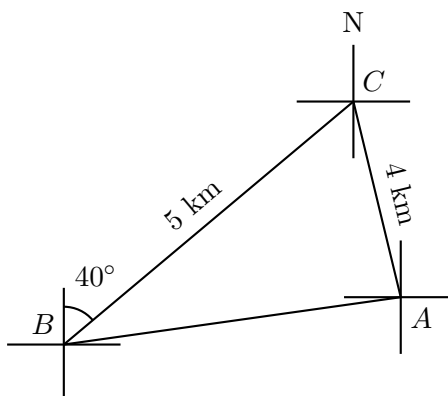
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Marks

- (a) Michael is training for a local marathon. He has trained by completing practice runs over the marathon course. So far he has completed three practice runs with times shown below.

Week	Time (hours)
1	3
2	2.7
3	2.43

- i. Show that these times form a geometric series with common ratio $r = 0.9$. **1**
 - ii. If this series continues, what would be his expected time in Week 5, completed to the nearest minute? **2**
 - iii. How many hours and minutes will he have run in total in his practice runs in these 5 weeks? **2**
 - iv. If the previous winning time for the marathon was 1 hour 15 min, how many weeks must he keep practising to be able to run the marathon in less than the previous winning time? **2**
- (b) A , B and C are markers in an orienteering course. $AC = 4$ km and $BC = 5$ km. The bearing of C from B is 040°T .



Copy or trace the diagram into your writing booklet.

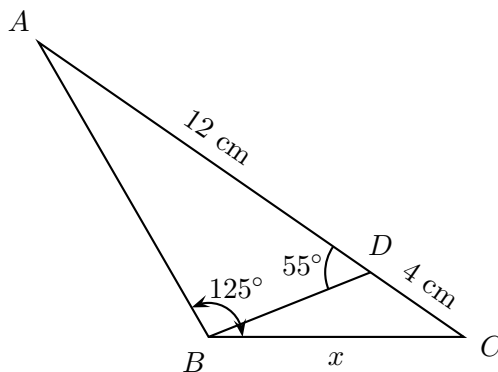
- i. If the bearing of B from A is 260°T , show that $\angle CBA = 40^\circ$, giving reasons. **2**
- ii. Find $\angle CAB$ to the nearest degree. **2**
- iii. Hence or otherwise, find the bearing of C from A . **1**

Question 5 (12 Marks)

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Marks

- (a) If $f(x) = x^2 - x$,
- Evaluate and expand $f(x + h)$. **1**
 - Hence or otherwise, differentiate $f(x) = x^2 - x$ from first principles. **3**
- (b) In $\triangle ABC$ as shown in the diagram, $\angle ABC = 125^\circ$, $\angle ADB = 55^\circ$, $AD = 12$ cm and $DC = 4$ cm.



NOT TO SCALE

- Show that $\triangle ABC \parallel \triangle BDC$. **3**
 - Find x , the length of BC . **2**
- (c) Let α and β be the solutions of $2x^2 - 6x - 1 = 0$.
- Find $\alpha + \beta$. **1**
 - Find $\alpha\beta$. **1**
 - Hence, find $3\alpha - \alpha^2$. **1**

Question 6 (12 Marks)

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Marks

- (a) A particle is moving in a straight line. Its velocity for $t \geq 0$ is given by

$$v = \frac{4}{t+1} - 2t$$

- i. Find when the particle changes direction. **2**
 - ii. Find the exact distance travelled in the first 2 seconds. **3**
- (b) For the function $y = x^3 - 3x^2 - 9x + 1$,
- i. Find the coordinates of any stationary points and determine their nature. **3**
 - ii. Find any points of inflexion. **2**
 - iii. Neatly sketch the curve. **2**

Question 7 (12 Marks)

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Marks

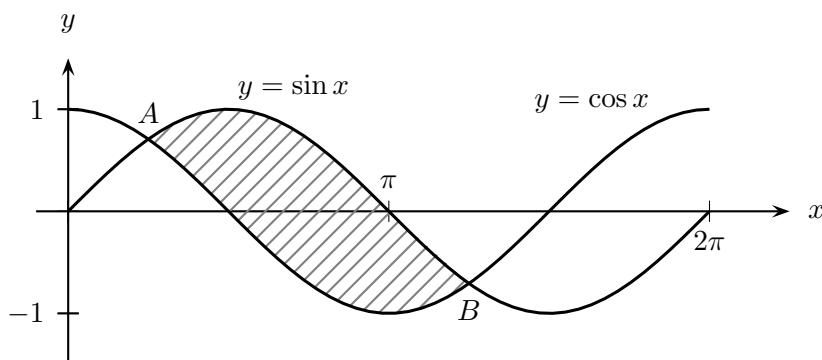
- (a) A farmer has a large tank full of water. The tank leaks water from a hole. The volume of water remaining in the tank, in litres, is given by

$$V = 4\,000 + 10\,000e^{-0.04t}$$

where t is the time in hours after the leakage commenced.

- i. How many litres of water were in the tank when the leakage commenced? **1**
- ii. At what rate is the water leaking after 5 hours? Answer correct to 1 decimal place. **2**
- iii. How many litres will eventually be in the tank after a long period of time? **1**
- iv. If the farmer realises the tank is leaking when the volume of water remaining is 6 000 L, how long did it take him to realise there was a hole in the tank? Answer correct to the nearest minute. **2**

- (b) The diagram shows the graphs $y = \sin x$ and $y = \cos x$, $0 \leq x \leq 2\pi$. The graphs intersect at A and B .



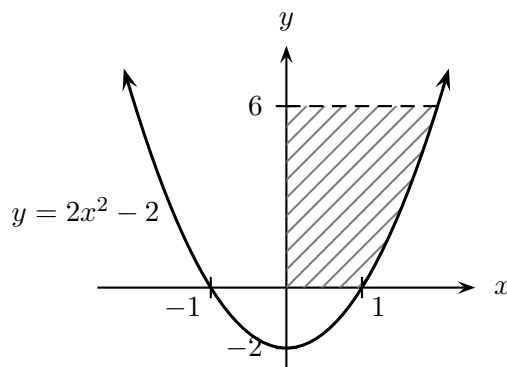
- i. Show that A has coordinates $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ and find the coordinates of B . **3**
- ii. Find the area enclosed by the two graphs. **3**

Question 8 (12 Marks)

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Marks

- (a) The diagram shows the region bounded by the curve $y = 2x^2 - 2$, the line $y = 6$ and the x and y axes. **3**



Find the volume of the solid of revolution formed when the region is rotated about the y axis.

- (b) Kevin plays computer games competitively. From past experience, Kevin has a 0.8 chance of winning a game of *Sawcraft* and a 0.6 chance of winning *CounterStrife*. During a LAN party he plays two games of *Sawcraft* and one of *CounterStrife*.

What is the probability that he will win:

- i. all 3 games? **1**
 - ii. No games? **2**
 - iii. At least 1 game? **2**
- (c) For the quadratic equation $x^2 + (p - 3)x - (2p + 1) = 0$,
- i. Show that the discriminant is $\Delta = p^2 + 2p + 13$. **2**
 - ii. Hence or otherwise, show that the quadratic equation $x^2 + (p - 3)x - (2p + 1) = 0$ will always have real, distinct roots for real valued p . **2**

Question 9 (12 Marks)

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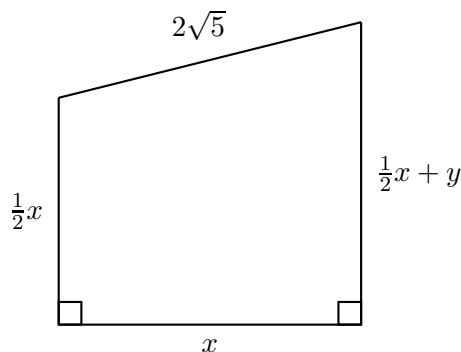
Marks

(a) Consider the geometric series $1 + \frac{4}{3}\sin^2 x + \frac{16}{9}\sin^4 x + \frac{64}{27}\sin^6 x + \dots$.

i. When the limiting sum exists, find its value in simplest form. **2**

ii. For what values of x in the interval $0 < x < \frac{\pi}{2}$ does the limiting sum of this series exist? **2**

(b) The diagram below represents (in metres) the dimensions of a small garden.



i. Show that $y = (20 - x^2)^{\frac{1}{2}}$. **2**

ii. Write an expression, in terms of x , for the perimeter P (in metres) of the garden, and find a value of x for which **4**

$$\frac{dP}{dx} = 0$$

iii. Establish whether this value of x gives a minimum or maximum value of P and find that value of P . **2**

Question 10 (12 Marks)

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Marks

- (a) A city has a growing population at a rate proportional to the current population, that is

$$\frac{dP}{dt} = kP$$

- i. Verify that $P(t) = P_0 e^{kt}$, $t > 0$ is a solution of the equation. **1**
 - ii. If the population on 1 January 2006, which is $t = 1$, was 147 200 and on 1 January 2007 (when $t = 2$) was 154 800, find the initial population and the value of k . Round your answer down to the nearest whole number. **2**
 - iii. Find the population on 1 January 2009. **1**
 - iv. Find the time it will take for the population to double. **2**
- (b) A car dealership has a car for sale for the cash price of \$20 000. It can also be purchased on terms over 3 years. The first 6 months are interest free. Subsequently, interest is charged at 12% per annum, calculated monthly. Repayments are to be made in equal monthly instalments at the end of the first month.

A customer purchases the car on these terms and agrees to monthly repayments of \$ M per month. Let \$ A_n be the amount owing at the end of the n -th month.

- i. Find an expression for A_6 . **1**
- ii. Show that $A_8 = (20\,000 - 6M)1.01^2 - M(1 + 1.01)$. **2**
- iii. Find an expression for A_{36} . **1**
- iv. Find the value of M . **2**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested marking scheme

Question 1

(a) (2 marks)

- ✓ [1] for correct value.
- ✓ [1] for 3 significant figures.

$$\left(\frac{1}{e^{2.5}} - 1\right) = 0.843 \text{ (3 s.f.)}$$

(b) (2 marks)

- ✓ [1] for multiplying by the fraction with appropriate conjugate surd.
- ✓ [1] for final answer.

$$\frac{\sqrt{2}}{1+\sqrt{5}} \times \frac{1-\sqrt{5}}{1-\sqrt{5}} = -\frac{\sqrt{2}-\sqrt{10}}{4} = \frac{\sqrt{10}-\sqrt{2}}{4}$$

(c) (2 marks)

- ✓ [1] for correct usage of chain rule.
- ✓ [1] for final answer.

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 3(4x+1)^2 \times 4 \\ &= 12(4x+1)^2 \end{aligned} \quad \left| \begin{array}{ll} y = u^3 & u = 4x+1 \\ y' = 3u^2 & u' = 4 \end{array} \right.$$

(d) (2 marks)

- ✓ [1] for correctly factorising xy .
- ✓ [1] for correctly factorising cubic.

$$\begin{aligned} x^4y - xy^4 &= xy(x^3 - y^3) \\ &= xy(x-y)(x^2 + xy + y^2) \end{aligned}$$

(e) i. (2 marks)

- ✓ [1] for identifying $32 = 2^5$.
- ✓ [1] for correct solution.

$$\begin{aligned} 2^{2x-3} &= 32 = 2^5 \\ 2x-3 &= 5 \\ 2x &= 8 \Rightarrow x = 4 \end{aligned}$$

ii. (2 marks)

- ✓ [1] for correctly factorising quadratic.
- ✓ [1] for $x = -1, 2$.

$$\begin{aligned} x^2 - x &= 2 \\ x^2 - x - 2 &= 0 \\ (x-2)(x+1) &= 0 \\ x &= -1, 2 \end{aligned}$$

Question 2

(a) i. (1 mark)

$$2x + 3y + 6 = 0$$

When $y = 0$, $2x + 6 = 0$

$$2x = -6 \Rightarrow x = -3$$

ii. (1 mark)

$$2x + 3y + 6 = 0$$

When $x = 0$, $3y + 6 = 0$

$$3y = -6 \Rightarrow y = -2$$

iii. (2 marks)

- ✓ [1] for using midpoint formula.
- ✓ [1] for final answer $C(3, -4)$.

$$\begin{aligned} (0, -2) &= \left(\frac{x_c + (-3)}{2}, \frac{y_c + 0}{2} \right) \\ \frac{x_c - 3}{2} &= 0 \quad \left| \quad \frac{y_c}{2} = -2 \right. \\ x_c &= 3 \quad \left| \quad y_c = -4 \right. \\ \therefore C(3, -4) \end{aligned}$$

iv. (2 marks)

- ✓ [1] for correct gradient of k .
- ✓ [1] for correct y intercept of k .

$$\begin{aligned} m_\ell &= -\frac{2}{3} \Rightarrow m_k = \frac{3}{2} \text{ as } \ell \perp k \\ y &= \frac{3}{2}x + b \Big|_{\substack{x=3 \\ y=-4}} \\ -4 &= \frac{3}{2} \times 3 + b \\ \therefore b &= -4 - \frac{9}{2} = -\frac{17}{2} \\ \therefore y &= \frac{3}{2}x - \frac{17}{2} \Rightarrow 3x - 2y - 17 = 0 \end{aligned}$$

v. (3 marks)

✓ [1] for each correct inequality.

$$\begin{cases} y \geq -\frac{2}{3}x - 2 \\ y \leq 0 \\ y \geq \frac{3}{2}x - \frac{17}{2} \end{cases} \quad \begin{cases} 2x + 3y + 6 \geq 0 \\ y \leq 0 \\ 3x - 2y - 17 \leq 0 \end{cases}$$

(b) (3 marks)

✓ [1] for application of product rule.

✓ [1] for finding function value at $x = e$.

✓ [1] for final answer.

$$\begin{aligned} y &= x^2 \ln x \\ u &= x^2 \quad v = \ln x \\ u' &= 2x \quad v' = \frac{1}{x} \\ \frac{dy}{dx} &= uv' + vu' = x^2 \cdot \frac{1}{x} + 2x \cdot \ln x \\ &= x + 2x \ln x \Big|_{x=e} \\ &= e + 2e \ln e = 3e \end{aligned}$$

The function value at $x = e$ is

$$y = x^2 \ln x \Big|_{x=e} = e^2$$

Substituting (e, e^2) into equation of the tangent,

$$\begin{aligned} \therefore y &= 3ex + b \Big|_{\substack{x=e \\ y=e^2}} \\ e^2 &= 3e^2 + b \\ b &= -2e^2 \\ \therefore y &= 3ex - 2e^2 \end{aligned}$$

Question 3

(a) i. (1 mark)

$$\begin{aligned} (x-2)^2 &= 4 \times 2(y+1) \\ \therefore a &= 2 \end{aligned}$$

ii. (1 mark)

$$\begin{aligned} V(2, -1) \quad a &= 2 \\ \therefore F(2, -1 + a) &= F(2, 1) \end{aligned}$$

iii. (1 mark)

$$\therefore y = -1 - a = -3$$

(b) i. (2 marks)

✓ [1] for correct differentiation of each term.

$$\frac{d}{dx} (2x^3 - x^{-1}) = 6x^2 + x^{-2}$$

ii. (1 mark)

✓ [1] for correct application of product or quotient rule

✓ [1] for correct final answer

$$\begin{aligned} \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\ &= \frac{e^{2x} \cos x - 2e^{2x} \sin x}{(e^{2x})^2} \quad \left| \begin{array}{l} u = \sin x \quad v = e^{2x} \\ u' = \cos x \quad v' = 2e^{2x} \end{array} \right. \\ &= \frac{\cancel{e^{2x}} (\cos x - 2 \sin x)}{\cancel{e^{2x}} \cdot e^{2x}} \\ &= \frac{\cos x - 2 \sin x}{e^{2x}} \end{aligned}$$

Alternatively, apply the product rule to $y = e^{-2x} \sin x$ to obtain

$$y' = e^{-2x} (\cos x - 2 \sin x)$$

(c) (2 marks)

✓ [1] for finding the primitive.

✓ [1] for correct evaluation of limits.

$$\begin{aligned} \int_1^e \left(x^2 + \frac{2}{x} \right) dx &= \frac{1}{3} x^3 + 2 \ln x \Big|_1^e \\ &= \frac{1}{3} (e^3 - 1) + 2(\ln e - \ln 1) \\ &= \frac{1}{3} e^3 + \frac{5}{3} \end{aligned}$$

(d) (3 marks)

✓ [1] recollection of Simpson's Rule

✓ [1] substitution of pronumerals.

✓ [1] evaluation.

$$\begin{aligned} A &\approx \frac{h}{3} (y_1 + 4 \sum y_{\text{even}} + 2 \sum y_{\text{odd}} + y_\ell) \\ &= \frac{\frac{1}{2}}{3} (12 + 4(8 + 3) + 0 + 5) \\ &= \frac{1}{6} (17 + 44) = \frac{61}{6} \end{aligned}$$

Question 4

(a) i. (1 mark)

$$\frac{T_2}{T_1} = \frac{2.7}{3.0} = 0.9 \quad \frac{T_3}{T_2} = \frac{2.43}{2.70} = 0.9$$

ii. (2 marks)

- ✓ [1] for substitution of pronumerals.
- ✓ [1] for correct answer to nearest minute.

$$T_n = ar^{n-1}$$

$$T_5 = 3 \times 0.9^4 = 1.9683 = 1 \text{ h } 59 \text{ min}$$

iii. (2 marks)

- ✓ [1] for substitution of pronumerals.
- ✓ [1] for correct answer to nearest minute.

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_5 = \frac{3 \times (0.9^5 - 1)}{0.9 - 1}$$

$$= 12.2853 = 12 \text{ h } 17 \text{ min}$$

iv. (2 marks)

- ✓ [1] for correct evaluation resulting in $n \approx 9.31$.
- ✓ [1] for correct answer to $\lceil n \rceil$.

$$T_n = 1.25 \text{ h} = \frac{3}{\div 3} \times 0.9^{n-1}$$

$$0.9^{n-1} = \frac{1.25}{3}$$

$$(n-1) \log 0.9 = \log \frac{1.25}{3}$$

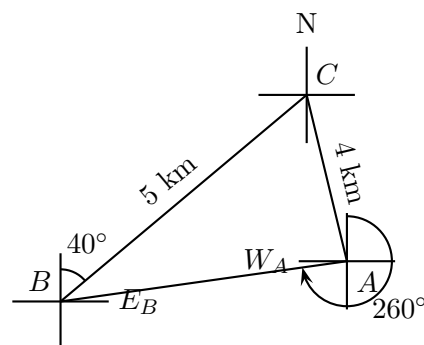
$$n = \frac{\log \frac{1.25}{3}}{\log 0.9} + 1 \approx 9.31$$

Michael must run for 10 weeks to improve on the previous record of 1 h 15 min.

(b) i. (2 marks)

- ✓ [1] for correct arithmetic.
- ✓ [1] for correct reasoning.

Any arithmetic/reasoning that is not acceptable will result in no marks awarded.



- $\angle W_A B = 10^\circ$ since the bearing of B from A is 260° .
- $\angle ABE_B = 10^\circ$ (alt. $\angle$ on $\parallel$ lines)
- $\therefore CBA = 90^\circ - 40^\circ - 10^\circ = 40^\circ$. (complementary $\angle$)

ii. (2 marks)

- ✓ [1] for application of sine rule.
- ✓ [1] for final answer.

$$\frac{\sin 40^\circ}{4} = \frac{\sin \angle CAB}{5}$$

$$\sin \angle CAB = \frac{5 \sin 40^\circ}{4} \approx 0.803$$

$$\angle CAB = 53^\circ 28' = 53^\circ \text{ (nearest } ^\circ \text{)}$$

iii. (1 mark)

The bearing of C from A is $53^\circ 28' + 260^\circ = 313^\circ 28'$. Also accept 313° .

Question 5

(a) i. (1 mark)

$$f(x+h) = (x+h)^2 - (x+h)$$

$$= x^2 + 2hx + h^2 - x - h$$

ii. (3 marks)

- ✓ [1] for recollection of limit.
- ✓ [1] for substitution.
- ✓ [1] for final answer.

$$\left| \begin{aligned} & f(x+h) - f(x) \\ &= (\cancel{x^2} + 2hx + h^2 - \cancel{x} - h) - (\cancel{x^2} - \cancel{x}) \\ &= 2xh + h^2 - h \end{aligned} \right|$$

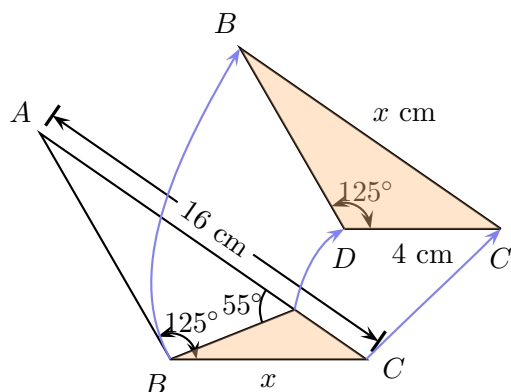
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(2x + h - 1)}{\cancel{h}} = 2x - 1$$

(b) i. (3 marks)

✓ [1] for each reason shown in summary.

In $\triangle ABC$ and $\triangle BDC$,1. $\angle CDB = 180^\circ - 55^\circ = 125^\circ = \angle ABC$ (supplementary $\angle$)2. $\angle BCD$ common to $\triangle ABC$ and $\triangle BDC$.3. $\therefore \angle DBC = \angle BAC$ (angle sum of $\triangle$) since two other pairs of angles are equal. $\therefore \triangle ABC \parallel \triangle BDC$ (AAA)

ii. (2 marks)

✓ [1] for relating corresponding sides in the same ratio.

✓ [1] for correctly evaluating x .Since $\therefore \triangle ABC \parallel \triangle BDC$, all corresponding sides are in the same ratio, i.e.

$$\frac{x}{4} = \frac{16}{x}$$

$$x^2 = 64 \Rightarrow x = 8$$

(c) i. (1 mark)

$$2x^2 - 6x - 1 = 0$$

$$\alpha + \beta = -\frac{b}{a} = -\frac{-6}{2} = 3 \quad (5.1)$$

ii. (1 mark)

$$\alpha\beta = \frac{c}{a} = -\frac{1}{2}$$

iii. (1 mark)

$$3\alpha - \alpha^2 = \alpha(3 - \alpha)$$

From (5.1), $\beta = 3 - \alpha$

$$\therefore \alpha(3 - \alpha) = \alpha\beta = -\frac{1}{2}$$

Question 6

(a) i. (2 marks)

✓ [1] for recollection of particle changing direction when $v = 0$.✓ [1] for correct arithmetic and reasoning to obtain $t = 1$.Particle changes direction when $v = 0$

$$\frac{4}{t+1} = 2t$$

$$4 = 2t(t+1)$$

$$2t^2 + 2t - 4 = 0$$

$$t^2 + t - 2 = 0$$

$$(t+2)(t-1) = 0$$

$$t = 1 \text{ since } t \geq 0$$

ii. (3 marks)

✓ [1] for applying absolute value to both terms of the distance.

✓ [1] for $d = |4 \ln 2 - 1| + |4 \ln \frac{3}{2} - 4|$.✓ [1] for $d = 4 \ln \frac{4}{3} + 3$ m.✓ [Note:] If $d = \int_0^2 \frac{4}{t+1} - 2t \, dt$ is used, a maximum of [1] mark is awarded.

$$d = \left| \int_0^1 \frac{4}{t+1} - 2t \, dt \right|$$

$$+ \left| \int_1^2 \frac{4}{t+1} - 2t \, dt \right|$$

$$= \left| 4 \ln(t+1) - t^2 \right|_0^1$$

$$+ \left| 4 \ln(t+1) - t^2 \right|_1^2$$

$$= \left| 4(\ln 2 - \ln 1) - (1^2 - 0^2) \right|$$

$$+ \left| 4(\ln 3 - \ln 2) - (2^2 - 1^2) \right|$$

$$= |4 \ln 2 - 1| + |4 \ln \frac{3}{2} - 3|$$

$$= (4 \ln 2 - 1) + (3 - 4 \ln \frac{3}{2})$$

$$= 4(\ln 2 - \ln \frac{3}{2}) + 2$$

$$= 4 \ln \frac{4}{3} + 2 \text{ m}$$

(b) i. (3 marks)

- ✓ [1] correct identification of $x = -1, 3$.
- ✓ [1] testing nature of stationary points.
- ✓ [1] finding coordinates & stating nature.

$$y = x^3 - 3x^2 - 9x + 1$$

$$\frac{dy}{dx} = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3)$$

$$= 3(x - 3)(x + 1)$$

Stationary pts. at $y' = 0$.

$$\therefore x = -1, 3$$

$$y = x^3 - 3x^2 - 9x + 1 \Big|_{x=-2} = -1$$

$$y = x^3 - 3x^2 - 9x + 1 \Big|_{x=-1} = 6$$

$$y = x^3 - 3x^2 - 9x + 1 \Big|_{x=0} = 1$$

$$y = x^3 - 3x^2 - 9x + 1 \Big|_{x=3} = -26$$

$$y = x^3 - 3x^2 - 9x + 1 \Big|_{x=4} = -19$$

x	-2	-1	0	3	4
$\frac{dy}{dx}$	+	0	-	0	+
y	-1	6		-26	-19

$\therefore (-1, 6)$ is a local maximum and $(3, -26)$ is a local minimum.

ii. (2 marks)

- ✓ [1] obtaining $y'' = 6x - 6$.
- ✓ [1] showing change in concavity when $x = 1$.

$$\frac{dy}{dx} = 3x^2 - 6x - 9 \Rightarrow \frac{d^2y}{dx^2} = 6x - 6$$

Pt. of inflexion when $y'' = 0$ & concavity change occurs, i.e.

$$6x - 6 = 0 \Rightarrow x = 1$$

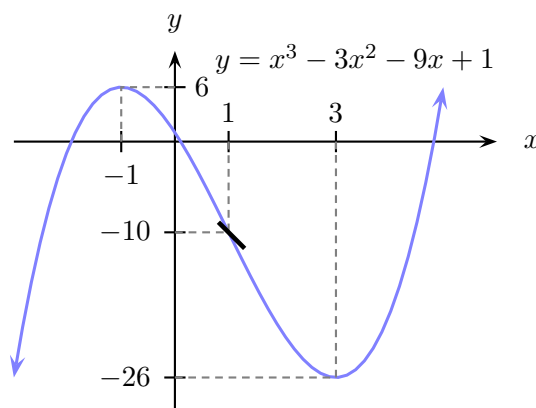
$$y'' = 6x - 6 \Big|_{x=0} < 0$$

$$y'' = 6x - 6 \Big|_{x=2} > 0$$

x	0	1	2
y''	−	0	+

iii. (2 marks)

- ✓ [1] shape of curve.
- ✓ [1] coords of stationary pts, pt. of inflexion.

**Question 7**

(a) i. (1 mark)

$$V(0) = 4\,000 + 10\,000e^0 = 14\,000 \text{ L}$$

ii. (2 marks)

- ✓ [1] for obtaining $V'(t) = -400e^{-0.04t}$.
- ✓ [1] for evaluating $V'(5) = -327.5 \text{ L/h}$.

$$V(t) = 4\,000 + 10\,000e^{-0.04t}$$

$$V'(t) = -0.04 \times 10\,000e^{-0.04t}$$

$$= -400e^{-0.04t}$$

$$V'(5) = -327.5 \text{ L/h (1 d.p.)}$$

iii. (1 mark)

$$\lim_{t \rightarrow \infty} (4\,000 + 10\,000e^{-0.04t}) = 4\,000 \text{ L}$$

iv. (2 marks)

- ✓ [1] for obtaining $e^{-0.04t} = \frac{1}{5}$.
- ✓ [1] for obtaining $t_1 = 40.23 = 40 \text{ h } 14 \text{ min}$.

Let t_1 be the time the farmer is aware of the leak.

$$V(t_1) = 6\,000 = 4\,000 + 10\,000e^{-0.04t}$$

$$\frac{2\,000}{10\,000} = \frac{10\,000e^{-0.04t}}{10\,000}$$

$$e^{-0.04t} = \frac{1}{5}$$

$$-0.04t = \ln \frac{1}{5}$$

$$t = \frac{\ln \frac{1}{5}}{-0.04} = 40.23 \dots \text{ h}$$

40 h 14 min have elapsed since the leak was discovered.

(b) i. (3 marks)

- ✓ [1] for solution of $\sin x = \cos x$.
 ✓ [2] for $A\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right), B\left(\frac{5\pi}{4}, -\frac{1}{\sqrt{2}}\right)$.

$$\begin{aligned}\sin x &= \cos x &\Rightarrow x &= \frac{\pi}{4}, \frac{5\pi}{4} \\ \tan x &= 1 \\ \therefore A\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right) & \quad B\left(\frac{5\pi}{4}, -\frac{1}{\sqrt{2}}\right)\end{aligned}$$

ii. (3 marks)

- ✓ [1] setting up integral.
 ✓ [1] for successfully finding primitive.
 ✓ [1] for $A = 2\sqrt{2}$.

$$\begin{aligned}A &= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \sin x \, dx - \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \cos x \, dx \\ &= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x) \, dx \\ &= -\cos x - \sin x \Big|_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \\ &= -\left(\cos \frac{5\pi}{4} - \cos \frac{\pi}{4}\right) - \left(\sin \frac{5\pi}{4} - \sin \frac{\pi}{4}\right) \\ &= -\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) \\ &= \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} = 2\sqrt{2}\end{aligned}$$

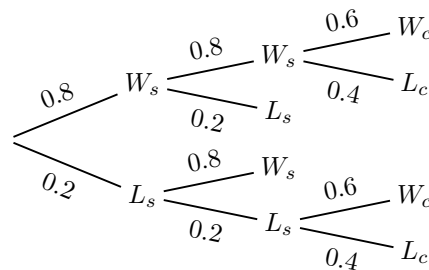
Question 8

(a) (3 marks)

- ✓ [1] for changing the subject to x^2 .
 ✓ [1] for setting up integral.
 ✓ [1] for solution $V = 15\pi$.

$$\begin{aligned}y &= 2x^2 - 2 \Rightarrow y + 2 = 2x^2 \\ x^2 &= \frac{y+2}{2} \\ V &= \pi \int x^2 \, dy \\ &= \pi \int_0^6 \frac{y+2}{2} \, dy \\ &= \frac{\pi}{2} \left(\frac{1}{2}y^2 + 2y \right) \Big|_0^6 \\ &= \frac{\pi}{2} \left(\frac{1}{2}(6^2) + 2(6) \right) \\ &= 15\pi \text{ units}^3\end{aligned}$$

(b) Tree diagram for this question:



i. (1 mark)

$$P(W_s W_s W_c) = (0.8)^2 \times (0.6) = 0.384$$

ii. (2 marks)

- ✓ [1] Find the complements of W_s, W_c :

$$P(L_s) = 1 - P(\overline{W_s}) = 0.2$$

$$P(L_c) = 1 - P(\overline{W_c}) = 0.4$$

- ✓ [1] correct evaluation of

$$P(L_s L_s L_c) = 0.016$$

$$P(L_s L_s L_c) = (0.2)^2 \times (0.4) = 0.016$$

iii. (2 marks)

- ✓ [1] Find the complement:

$$P(\text{win at least 1}) = 1 - P(L_s L_s L_c)$$

- ✓ [1] correct evaluation.

$$\begin{aligned}P(\text{win at least 1}) &= 1 - P(\text{win none}) \\ &= 1 - P(L_s L_s L_c) \\ &= 1 - 0.016 = 0.984\end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned}a &= 1 \quad b = (p-3) \quad c = -(2p+1) \\ \Delta &= b^2 - 4ac \\ &= (p-3)^2 - 4 \times 1 \times -(2p+1) \\ &= p^2 - 6p + 9 + 8p + 4 \\ &= p^2 + 2p + 13\end{aligned}$$

ii. (2 marks)

- ✓ [1] correctly evaluating $\Delta < 0$.
 ✓ [1] logical reasoning.

$$\begin{aligned}\Delta &= 2^2 - 4 \times 1 \times 13 = 4 - 42 < 0 \\ \therefore \Delta &= p^2 + 2p + 13 > 0 \forall p\end{aligned}$$

Since $\Delta > 0$, therefore the quadratic always has real, distinct roots for real values of p .

Question 9

(a) i. (2 marks)

- ✓ [1] for correct identification $a = 1$,
 $r = \frac{4}{3} \sin^2 x$.
 ✓ [1] for correct substitution of a and r
 to the limiting sum $S = \frac{1}{1 - \frac{4}{3} \sin^2 x}$.

$$\begin{aligned} a &= 1 & r &= \frac{4}{3} \sin^2 x \\ S &= \frac{a}{1-r} = \frac{1}{1 - \frac{4}{3} \sin^2 x} \times \frac{3}{3} \\ &= \frac{3}{3 - 4 \sin^2 x} \end{aligned}$$

ii. (2 marks)

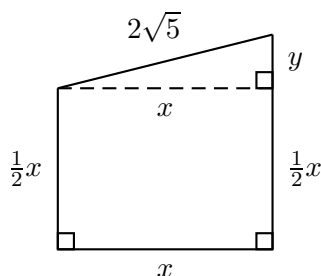
- ✓ [1] for equating $0 < \left| \frac{4}{3} \sin^2 x \right| < 1$.
 ✓ [1] for finding $0 < x < \frac{\pi}{3}$.

 $|r| < 1$ for limiting sum to exist.

$$\begin{aligned} \left| \frac{4}{3} \sin^2 x \right| &< 1 \\ 0 &< \frac{4}{3} \sin^2 x < 1 \\ 0 &< \sin^2 x < \frac{3}{4} \\ 0 &< \sin x < \frac{\sqrt{3}}{2} \\ 0 &< x < \frac{\pi}{3} \end{aligned}$$

(b) i. (2 marks)

- ✓ [1] for using Pythagoras' Theorem
 ✓ [1] for showing required $y = (20 - x^2)^{\frac{1}{2}}$



$$\begin{aligned} x^2 + y^2 &= (2\sqrt{5})^2 \\ x^2 + y^2 &= 4 \times 5 = 20 \\ y^2 &= 20 - x^2 \Rightarrow y = (20 - x^2)^{\frac{1}{2}} \end{aligned}$$

ii. (4 marks)

- ✓ [1] for obtaining

$$P = 2x + (20 - x^2)^{\frac{1}{2}} + 2\sqrt{5}$$

- ✓ [1] differentiating $P(x)$ correctly.
 ✓ [1] obtaining $x^2 = 16$.
 ✓ [1] concluding $x = 4$ as x is a length.

$$\begin{aligned} P &= x + \frac{1}{2}x + \left(\frac{1}{2}x + y\right) + 2\sqrt{5} \\ &= 2x + (20 - x^2)^{\frac{1}{2}} + 2\sqrt{5} \\ \frac{dP}{dx} &= 2 + \frac{1}{2} \times (-2x) \times (20 - x^2)^{-\frac{1}{2}} \\ &= 2 - \frac{x}{\sqrt{20 - x^2}} = 0 \\ \frac{x^2}{20 - x^2} &= 4 \\ x^2 &= 80 - 4x^2 \\ 5x^2 &= 80 \Rightarrow x^2 = 16 \\ \div 5 & \quad \div 5 \end{aligned}$$

 $\therefore x = 4$ since x is a length & $x > 0$

iii. (2 marks)

- ✓ [1] Checking $\frac{dP}{dx}$ within a neighbourhood of $x = 4$.
 ✓ [1] Evaluating $P(4) = 2\sqrt{5} + 10$.

$$\begin{aligned} \frac{dP}{dx} &\text{ exists when } x \geq 0 \text{ and } x \leq \sqrt{20}. \\ \frac{dP}{dx} &= 2 - \frac{x}{\sqrt{20 - x^2}} \Big|_{x=3} = 1.095 \\ \frac{dP}{dx} &= 2 - \frac{x}{\sqrt{20 - x^2}} \Big|_{x=\sqrt{17}} = -0.38 \end{aligned}$$

x	3	4	$\sqrt{17}$
$\frac{dy}{dx}$	+	0	-
y	13.8	$2\sqrt{5} + 10$	14.5

$\therefore x = 4$ makes $P(4) = 2\sqrt{5} + 10$ a local maximum.

Question 10

(a) i. (1 mark)

Differentiating $P(t)$,

$$P'(t) = P_0 k e^{kt} = k \overbrace{P_0 e^{kt}}^{=P(t)} = kP$$

ii. (2 marks)

✓ [1] for obtaining $k = \ln \frac{387}{368}$.✓ [1] for obtaining $P_0 = 139\,973$.

$$P(1) = 147\,200 = P_0 e^k \quad (10.1)$$

$$P(2) = 154\,800 = P_0 e^{2k} \quad (10.2)$$

(10.2) $\div$ (10.1):

$$\frac{154\,800}{147\,200} = e^k$$

$$k = \ln \frac{387}{368} \quad (10.3)$$

Substitute (10.3) to (10.1)

$$147\,200 = P_0 \exp\left(\ln \frac{387}{368}\right) = P_0 \times \frac{387}{368}$$

$$P_0 = \frac{147\,200 \times 368}{387} = 139\,973$$

iii. (1 mark)

$$P(4) = 139\,973 \exp\left(\ln \frac{387}{368} \times 4\right)$$

$$= 171\,197$$

iv. (2 marks)

✓ [1] for obtaining $\ln 2 = t \ln \frac{387}{368}$.✓ [1] for obtaining $t \approx 14$ years.

$$P(t) = 2P_0 = P_0 \exp\left(\ln \frac{387}{368} \times t\right)$$

$$\ln 2 = t \ln \frac{387}{368}$$

$$t = \frac{\ln 2}{\ln \frac{387}{368}}$$

$$= 13.7688 \dots$$

$$= 14 \text{ years}$$

(b) i. (1 mark)

Since no interest is applied in the first 6 mths, then the amount owing will be

$$A_1 = 20\,000 - M$$

$$A_2 = 20\,000 - 2M$$

$$\vdots$$

$$A_6 = 20\,000 - 6M$$

ii. (2 marks)

✓ [1] for correctly finding A_7 .✓ [1] for correctly finding A_8 .After the 6th month, 12% p.a. = 0.01 p.m. interest is applied to A_6 .

$$A_7 = A_6 \times 1.01 - M$$

$$= (20\,000 - 6M) \times 1.01 - M$$

$$A_8 = A_7 \times 1.01 - M$$

$$= ((20\,000 - 6M) \times 1.01 - M) \times 1.01 - M$$

$$= (20\,000 - 6M) \times 1.01^2$$

$$- 1.01M - M$$

$$= (20\,000 - 6M) \times 1.01^2 - M(1 + 1.01)$$

iii. (1 mark)

$$A_9 = (20\,000 - 6M) \times 1.01^3$$

$$- M(1 + 1.01 + 1.01^2)$$

$$A_{36} = (20\,000 - 6M) \times 1.01^{30}$$

$$- M(\underbrace{1 + 1.01 + 1.01^2 + \dots + 1.01^{29}}_{30 \text{ terms}})$$

iv. (2 marks)

✓ [1] for finding $S_{30} = 100(1.01^{30} - 1)$.✓ [1] for finding $M = \$628.78$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{30} = \frac{1(1.01^{30} - 1)}{1.01 - 1} = 100(1.01^{30} - 1)$$

 $A_{36} = 0$ as the loan is repaid and no amount is outstanding.Letting $K = 1.01^{30}$,

$$(20\,000 - 6M) \times K = 100M(K - 1)$$

$$20\,000K - 6MK = 100M(K - 1)$$

$$20\,000K = 100M(K - 1) + 6MK$$

$$= M(100K - 100 + 6K)$$

$$= M(106K - 100)$$

$$\therefore M = \frac{20\,000 \times 1.01^{30}}{106 \times 1.01^{30} - 100} = 628.78$$

Their repayment is \$628.78 per month.